Robert Wissing

Curriculum Vitae

Thereses gate 28C, 0168 Oslo
Contact: +4794285790
robertwi@astro.uio.no

Key Qualifications

PhD degree in astrophysics with 15 published scientific articles within the topics of numerical magneto-hydrodynamics, dynamo processes, galaxy formation, planetary impacts and equation of state.

Employement

2023-Present **University of Oslo**, Oslo, Norway
Postdoctoral Researcher - Head of ChaNGa code for SPACE CoE

Education

2022 **University of Oslo**, Oslo, Norway
Ph.D. in Astrophysics, Supervisor: *Dr. Sijing Shen*
Research Thesis: Simulating Galactic Dynamo Processes with Smoothed Particle Magnetohydrodynamics
Award: Yara's Birkeland Prize 2023

2017 **Lund University**, Lund, Sweden
M.Sc. in Astrophysics, Supervisor: *Dr. David Hobbs*
Research Thesis: Simulations of the lunar-forming impact using a new equation of state

2015 **Lund University**, Lund, Sweden
B.Sc. in Astrophysics, Supervisor: *Dr. Ross Church*
Research Thesis: Common Envelope Evolution

Students mentored

2026–Present **Simon Saetre**, Oslo University
Thesis: Magnetic Fields in Cosmological Zoom-In Simulations

2025 – 2026 **Levente Orha**, Oslo University
Thesis: The effect of different numerical SPH model for giant-impact simulations

2025 – 2026 **Henrik Sande**, Oslo University

Thesis: Evaluating the role of equation of state and initial conditions in giant impact simulations

2024–Present **Joel Immanuel**, Oslo University

Thesis: The Role of Magnetic Fields in Galaxy Mergers

Teaching Experience

2023 – 2026 **Teacher**, Oslo University

Special Curriculum: ASTSP100 - Numerical modeling of magnetic fields in galaxy formation

2023 – 2026 **Teacher**, Oslo University

Special Curriculum: ASTSP100 - Numerical modeling of planetary impacts

2017 – 2026 **Teaching Assistant**, Oslo University

Course: AST4320 Cosmology and Extragalactic Astronomy

2017 – 2021 **Guest Lecturer**, Oslo University

Course: AST4320 Cosmology and Extragalactic Astronomy

Topic: A taste of numerical hydrodynamics

2017 **Course material**, Lund University

Course: ASTM22 Computational Astrophysics

Material: Planetary models and initial condition files

Proposals

EuroHPC - Development Call

2026 – 2027 SPACE ChaNGa branch - Final development and testing.

2024 – 2025 SPACE Changa branch - Development and Optimization-Deucalion

2024 – 2025 SPACE Changa branch - Development and Optimization - MareNostrum

2024 – 2025 SPACE Changa branch - Development and Optimization

EuroHPC - Benchmark Call

2023 SPACE - Changa branch - Benchmark Karolina GPU + LUMI

2023 SPACE - Changa branch

ALMA Cycle 11 Proposal 2024.1.00750.S - Co-investigator

2024 ALMA Cycle 11 Dust polarization in a dual AGN host galaxy undergoing powerful feedback

Approved with Grade A priority; observations pending.

International Conferences and Seminar Talks

Invited talks

2023 **Birkeland prize lecture**, Oslo, Norway

Simulating Galactic Dynamo Processes with Smoothed Particle Magnetohydrodynamics

2019 **University of Zurich, Institute for Computational Science** , Zurich, Switzerland

Smoothed particle magnetohydrodynamics with the geometric density average force expression

2018 **University of Zurich, Institute for Computational Science** , Zurich, Switzerland

A new equation of state for planetary impacts

Contributed talks

2026 **SPACE General Assembly Heraklion**

Talk: The ChaNGa code

2025 **SPACE Winter School in Sesto Pusteria**

Tutorial + Talk: The ChaNGa code

2025 **SPACE 2nd review meeting Luxemburg**

Talk: The ChaNGa code

2025 **International Workshop on Galaxy Formation + AGORA in Asia 2025**

Talk: Recent Improvements in SPMHD modeling of galaxies

2025 **N-Body Shop Excellence Conference 2025**

Talk: Space Project and Improvements to SPMHD

2025 **SPACE General Assembly Barcelona**

Talk: The ChaNGa code

2024 **SPACE COE training**

Talk: The ChaNGa code and the Charm++ framework

2024 **SPACE review meeting Luxemburg**

Talk: The ChaNGa code

2024 **SPACE General Assembly Munich**

Talk: The ChaNGa code

2024 **UiO Colloquium**

Talk: The ChaNGa code and computational challenges for cosmological codes

2022 **N-body Workshop 2022**

Talk: Numerical dependencies of the galactic dynamo in isolated galaxies with SPH

2021 **N-body Virtual Workshop 2021**
Talk: Magnetorotational instability in SPH

2018 **Santa Cruz Galaxy Workshop 2018**
Talk: Smoothed particle magnetohydrodynamics with the geometric density average force expression

Computational Skills

Programming: Python, Fortran, C, C++, Matlab, bash, Java, C#, php, html

OS: UNIX, macOS, Windows

Software: Photoshop, Excel, Word, PowerPoint

Hydrodynamic Codes: Gasoline, ChaNGa, phantom, vine

Analysis Tools: Splash, Paraview, Pynbody, Tipsy

Other Work Experience

2010 – 2012 **Pikaia Games**, *Game Designer & Writer*.
Project: Renations

2009 – 2011 **Compless**, *Computer Assembler*, Mjölby, Sweden.

Other Skills

Language: Swedish(native), English(fluent), Spanish(basic), Russian(basic) and Japanese(basic)

References

Name: Professor Sijing Shen
Connection: PhD Supervisor
Workplace: Institute of Theoretical Astrophysics, University of Oslo
Phone: +47 22856547
email: sijing.shen@astro.uio.no

Name: Professor James Wadsley
Connection: Collaborator
Workplace: Department of Astronomy, McMaster University
Phone: +1 905 525 9140 x 27106
email: wadsley@mcmaster.ca

NAME: Professor Tom Quinn
CONNECTION: Collaborator
WORKPLACE: Department of Astrophysics, University of Washington
PHONE:+1 206 685 9009
EMAIL: trq@astro.washington.edu

Publication List

See ADS for full publication list.

[1] Improved volume partition for smoothed particle hydrodynamics
R. Wissing & J. Wadsley (Submitted to A&A).

[2] A multi-scale view of molecular gas and magnetic fields in the Antennae galaxies
A. Lasrado et al. (incl. **R. Wissing**) 2026 (Submitted to A&A), arXiv:2610.03346.
Contribution: Discussion and revision of the manuscript.

[3] The seven dwarfs illuminated. The impact of radiation on dwarf galaxies and their circumgalactic medium
B. Baumschlager et al. (incl. **R. Wissing**) 2026, A&A, 712, A99.
Contribution: Discussion and revision of the manuscript.

[4] Fast, Stable, and Physical: Hyperbolic, Magnetic Field-Aligned Diffusion in SPH
N. Owens, J. Wadsley, **R. Wissing** and B. Keller 2026 (Preprint), arXiv:2604.21983.
Contribution: Provided the underlying SPMHD implementation, helped in the generation of the initial conditions and contributed to discussion and revising of the manuscript.

[5] Optimized smoothing kernels for smoothed particle hydrodynamics
R. Wissing, T. Quinn, B. Keller, J. Wadsley and S. Shen 2026, A&A, 706, A73.

[6] The LISA Astrophysics "Disc-IMRI" Code Comparison Project: Intermediate-Mass-Ratio Binaries in AGN-Like Discs
A. Derdzinski et al. (incl. **R. Wissing**) 2025 (Submitted to the Open Journal of Astrophysics), arXiv:2512.10893.
Contribution: Implemented the physical viscosity prescription and local isothermal equation of state into Gasoline2. Verification tests of the physical viscosity, modeling the Taylor-green vortex and the spreading disk case. Conversion of the initial conditions to tipsy format, and contributed to discussion and revising of the manuscript.

[7] In-Situ High Performance Visualization for Astronomy & Cosmology
N. Tuccari et al. (incl. **R. Wissing**) 2025, ADASS XXXIV conference contribution, arXiv:2510.24547.
Contribution: Responsible for the ChaNGa part of the SPACE CoE project. Helped to couple the in-situ visualization pipeline to ChaNGa and provided simulation setups and test data.

[8] EuroHPC SPACE CoE: Redesigning Scalable Parallel Astrophysical Codes for Exascale
N. Shukla et al. (incl. **R. Wissing**) 2025, CF '25 Companion, 177–184.
Contribution: Responsible for the ChaNGa part of the SPACE CoE project.

[9] High Performance Visualization for Astrophysics and Cosmology
N. Tuccari et al. (incl. **R. Wissing**) 2025, Proceedings of the 33rd Euromicro International Conference on Parallel, Distributed, and Network-Based Processing (PDP), 443–450.
Contribution: Provided ChaNGa simulation data and input on visualization techniques of SPH data.

[10] Numerical dependencies of the galactic dynamo in isolated galaxies with SPH
R. Wissing & S. Shen 2023, A&A, 673, A47.

[11] Axisymmetric smoothed particle magnetohydrodynamics
D. García-Senz, **R. Wissing**, R. M. Cabezón, E. Vurgun and M. Linares 2023, MNRAS, 518, 4115–4131.
Contribution: Performed the reference SPMHD simulations with Gasoline2 against which the axisymmetric implementation was compared, provided test problems, produced several of the figures, and contributed to revising the manuscript.

[12] Magnetorotational instability with smoothed particle hydrodynamics
R. Wissing, S. Shen, J. Wadsley and T. Quinn 2022, A&A, 659, A91.

[13] Smoothed particle magnetohydrodynamics with the geometric density average force expression
R. Wissing & S. Shen 2020, A&A, 638, A140.

[14] A new equation of state applied to planetary impacts. II. Lunar-forming impact simulations with a primordial magma ocean
R. Wissing & D. Hobbs 2020, A&A, 643, A40.

[15] A new equation of state applied to planetary impacts. I. Models of planetary interiors
R. Wissing & D. Hobbs 2020, A&A, 635, A21.